

Finding 6: Reasoning fails to emerge without meticulous reward design

While symbolic reasoning can emerge in simple, single-turn tasks under weak supervision, it fails to persist in multi-turn environments without the reward design explicitly encouraging interpretable intermediate reasoning steps. We observe that even with structured prompts, reasoning gradually decays during training if the reward signal focuses only on final outcomes. This suggests that without meticulous reward shaping, agents may tend to collapse into shortcut behaviors that bypass reasoning altogether.